

ECON 320 — Mathematical Economics

Integration and Economic Dynamics

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Primary Text A. C. Chiang & K. Wainwright, *Fundamental Methods of Mathematical Economics*, 4th ed., McGraw-Hill, 2005.

Supplementary Hoy et al., *Mathematics for Economics*, 3rd ed., MIT Press, 2011.
Simon & Blume, *Mathematics for Economists*, Norton, 1994.

Topics Indefinite and definite integrals, rules of integration, improper integrals, consumer and producer surplus, present value of income streams, first-order differential equations, phase diagrams, stability analysis, the Solow growth model, first-order difference equations.

Contents

1	Introduction: Integration as Accumulation	3
2	The Indefinite Integral	3
2.1	Definition and Antiderivatives	3
2.2	Basic Integration Rules	4
2.3	Techniques of Integration	5
3	The Definite Integral	6
3.1	Definition and the Fundamental Theorem	6
3.2	Improper Integrals	7
4	Economic Applications of Integration	8
4.1	Consumer Surplus and Producer Surplus	8
4.2	Present Value of Continuous Income Streams	9
4.3	Capital Accumulation	10
5	First-Order Differential Equations	11
5.1	Definitions and Classification	11
5.2	Linear First-Order ODEs	11
5.3	Separable Differential Equations	13
6	Phase Diagrams and Stability Analysis	14
6.1	Phase Diagrams for Autonomous Equations	14
6.2	Stability Conditions	15
7	First-Order Linear Difference Equations	16
7.1	Definition and Solution	16
8	Economic Dynamics: Selected Applications	18
8.1	The Domar Growth Model	18
8.2	Continuous-Time Discounting and Optimal Control Preview	18
9	Practice Questions	19
9.1	Section A: Integration Techniques	19
9.2	Section B: Differential Equations	20
9.3	Section C: Difference Equations	21
10	Full Solutions	22
10.1	Solutions: Section A	22
10.2	Solutions: Section B	24
10.3	Solutions: Section C	26
	Summary: Key Results	28

1 Introduction: Integration as Accumulation

Differentiation measures *rates of change*; integration measures *accumulation*. These two operations are inverse to each other — a relationship made precise by the Fundamental Theorem of Calculus.

In economics, integration arises in three distinct but related contexts:

- **Area under a curve:** Consumer surplus, producer surplus, deadweight loss, and the Lorenz curve all involve areas under or between curves.
- **Aggregation over time:** Present values of continuous income streams, total capital accumulation, and cumulative production are integrals over time.
- **Solving differential equations:** Dynamic models — the Solow growth model, the Ramsey–Cass–Koopmans model, price adjustment dynamics — require integrating differential equations to obtain time paths.

Economic *dynamics* studies how endogenous variables evolve over *time* in response to shocks, policies, or intrinsic forces. Static analysis answers “what is the equilibrium?”; dynamic analysis answers “how does the economy get there, and will it arrive?”

2 The Indefinite Integral

2.1 Definition and Antiderivatives

Definition Indefinite Integral (Antiderivative)

A function $F(x)$ is an **antiderivative** of $f(x)$ on an interval if $F'(x) = f(x)$ for all x in that interval.

The **indefinite integral** of $f(x)$ is the family of all antiderivatives:

$$\int f(x) dx = F(x) + C$$

where $C \in \mathbb{R}$ is an arbitrary constant of integration. The function $f(x)$ is called the **integrand**.

Since any two antiderivatives differ by a constant, the indefinite integral represents a *family* of functions. The constant C is determined by an **initial condition** — a known value of F at some point.

2.2 Basic Integration Rules

Theorem Standard Integration Formulae

Let f and g be integrable and a, n be constants ($n \neq -1$):

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (\text{Power Rule})$$

$$\int \frac{1}{x} dx = \ln |x| + C \quad (\text{Reciprocal})$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C \quad (\text{Exponential})$$

$$\int a^x dx = \frac{a^x}{\ln a} + C, \quad a > 0, a \neq 1 \quad (\text{General Exp.})$$

$$\int [af(x) + bg(x)] dx = a \int f(x) dx + b \int g(x) dx \quad (\text{Linearity})$$

$$\int e^x dx = e^x + C \quad (\text{Natural Exp.})$$

$$\int \ln x dx = x \ln x - x + C, \quad x > 0 \quad (\text{Natural Log})$$

(Chiang & Wainwright, p. 407; Simon & Blume, pp. 540–545)

Worked Example Applying the Basic Rules

- $\int (3x^4 - 5x^2 + 7x - 2) dx = \frac{3x^5}{5} - \frac{5x^3}{3} + \frac{7x^2}{2} - 2x + C$
- $\int \frac{4}{x^3} dx = 4 \int x^{-3} dx = 4 \cdot \frac{x^{-2}}{-2} + C = -\frac{2}{x^2} + C$
- Find $TC(Q)$ given $MC(Q) = 3Q^2 - 8Q + 5$ and $TC(0) = 100$ (fixed cost):

$$TC(Q) = \int (3Q^2 - 8Q + 5) dQ = Q^3 - 4Q^2 + 5Q + C$$

Applying $TC(0) = 100$: $C = 100$. So $TC(Q) = Q^3 - 4Q^2 + 5Q + 100$.

4. Economic interpretation: $K(t) = \int I(t) dt$ — the capital stock at time t is the cumulative integral of investment flows. If $I(t) = I_0 e^{gt}$, then $K(t) = (I_0/g)e^{gt} + C$, and the constant C is determined by the initial capital stock $K(0) = K_0$.

2.3 Techniques of Integration

Integration by Substitution

Theorem Substitution Rule

If $u = g(x)$ is differentiable and f is continuous, then:

$$\int f(g(x)) g'(x) dx = \int f(u) du$$

Procedure: identify an inner function $u = g(x)$, compute $du = g'(x) dx$, substitute, integrate in u , substitute back.

Worked Example Substitution

1. $\int 2x(x^2 + 3)^5 dx$:

Let $u = x^2 + 3$, $du = 2x dx$: $\int u^5 du = \frac{u^6}{6} + C = \frac{(x^2 + 3)^6}{6} + C$

2. $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$:

Let $u = \sqrt{x}$, $du = dx/(2\sqrt{x})$, so $dx/\sqrt{x} = 2 du$: $\int e^u \cdot 2 du = 2e^u + C = 2e^{\sqrt{x}} + C$

3. $\int \frac{x}{x^2 + 1} dx$:

Let $u = x^2 + 1$, $du = 2x dx$: $\frac{1}{2} \int \frac{du}{u} = \frac{1}{2} \ln(x^2 + 1) + C$

Integration by Parts

Theorem Integration by Parts

$$\int u dv = uv - \int v du$$

Choose u and dv so that $\int v du$ is easier than the original. A useful mnemonic for choosing u : **LIATE** (Logarithm, Inverse trig, Algebraic, Trig, Exponential — choose u from whichever comes first). (*Chiang & Wainwright, p. 411*)

Worked Example Integration by Parts

1. $\int xe^x dx$:

Let $u = x$, $dv = e^x dx$; then $du = dx$, $v = e^x$:

$$\int xe^x dx = xe^x - \int e^x dx = xe^x - e^x + C = e^x(x - 1) + C$$

2. $\int x^2 e^{-x} dx$:

Apply parts twice. Let $u = x^2$, $dv = e^{-x}dx$; $du = 2x dx$, $v = -e^{-x}$:

$$= -x^2e^{-x} + 2 \int xe^{-x} dx$$

Apply parts again to $\int xe^{-x} dx$: $u = x$, $dv = e^{-x}dx$; $v = -e^{-x}$:

$$= -x^2e^{-x} + 2 \left[-xe^{-x} + \int e^{-x} dx \right] = -x^2e^{-x} - 2xe^{-x} - 2e^{-x} + C = -e^{-x}(x^2 + 2x + 2) + C$$

3. $\int \ln x dx$:

Let $u = \ln x$, $dv = dx$; $du = dx/x$, $v = x$:

$$\int \ln x dx = x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - x + C$$

Economic application: The present value of a linearly growing payment $\int_0^T t e^{-rt} dt$ requires integration by parts twice; this is the *present value of a growing annuity*.

3 The Definite Integral

3.1 Definition and the Fundamental Theorem

Definition Definite Integral

The **definite integral** of f from a to b is:

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x = F(b) - F(a)$$

where F is any antiderivative of f and the limit is the Riemann sum. Geometrically, $\int_a^b f(x) dx$ equals the *net signed area* between the curve $y = f(x)$ and the x -axis over $[a, b]$.

Theorem Fundamental Theorem of Calculus

Let f be continuous on $[a, b]$ and F an antiderivative of f .

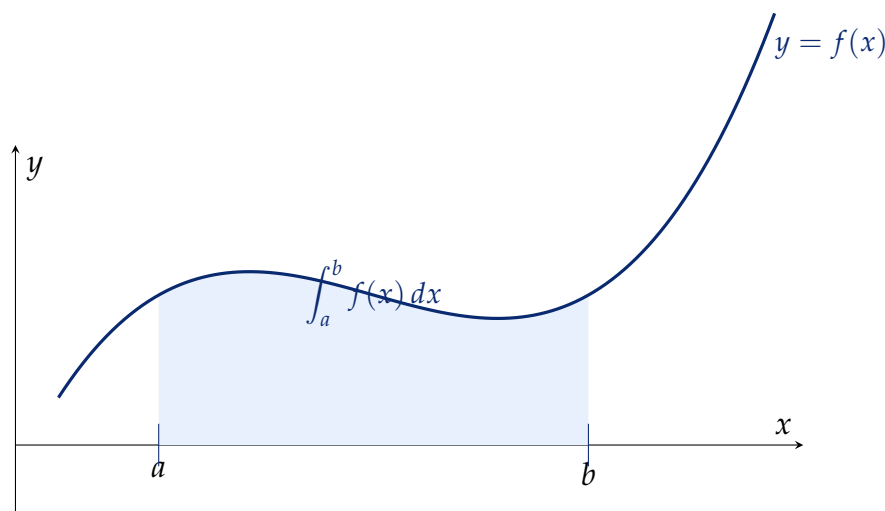
1. **Part I** (Differentiation of integrals): If $G(x) = \int_a^x f(t) dt$, then $G'(x) = f(x)$.

2. **Part II** (Evaluation): $\int_a^b f(x) dx = F(b) - F(a) \equiv F(x)|_a^b$.

(Chiang & Wainwright, p. 417; Simon & Blume, p. 550)

Properties of the Definite Integral

- $\int_a^a f(x) dx = 0$
- $\int_a^b f(x) dx = -\int_b^a f(x) dx$
- $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$ (additivity)
- $\int_a^b [af(x) + bg(x)] dx = a \int_a^b f dx + b \int_a^b g dx$ (linearity)
- If $f(x) \geq g(x)$ on $[a, b]$: $\int_a^b f dx \geq \int_a^b g dx$



The definite integral equals the net signed area between $f(x)$ and the x -axis over $[a, b]$.

3.2 Improper Integrals

Definition Improper Integrals

An integral is **improper** when the interval is unbounded or the integrand has a discontinuity within the interval.

Type I (infinite limits):

$$\int_a^\infty f(x) dx = \lim_{T \rightarrow \infty} \int_a^T f(x) dx$$

The integral **converges** if this limit is finite; otherwise it **diverges**.

Type II (discontinuous integrand): similarly handled with limits.

Worked Example Improper Integrals in Economics

1. **Perpetuity:** $\int_0^{\infty} Ce^{-rt} dt = -\frac{C}{r}e^{-rt} \Big|_0^{\infty} = 0 - \left(-\frac{C}{r}\right) = \frac{C}{r}$

(Requires $r > 0$.) This is the present value of a constant income stream C in perpetuity, discounted continuously at rate r .

2. **Growing perpetuity:** $\int_0^{\infty} Ce^{gt}e^{-rt} dt = C \int_0^{\infty} e^{-(r-g)t} dt = \frac{C}{r-g}$, provided $r > g$.

This is the Gordon Growth Model in continuous time: present value of a stream growing at rate g and discounted at rate $r > g$.

3. **Divergence when $r \leq g$:** If $r \leq g$ the integral diverges: the present value is infinite because the income stream grows at least as fast as the discount rate.

4 Economic Applications of Integration

4.1 Consumer Surplus and Producer Surplus

Definition Consumer and Producer Surplus

Let $P = D(Q)$ be the inverse demand (willingness-to-pay) and $P = S(Q)$ be the inverse supply, with equilibrium (Q^*, P^*) .

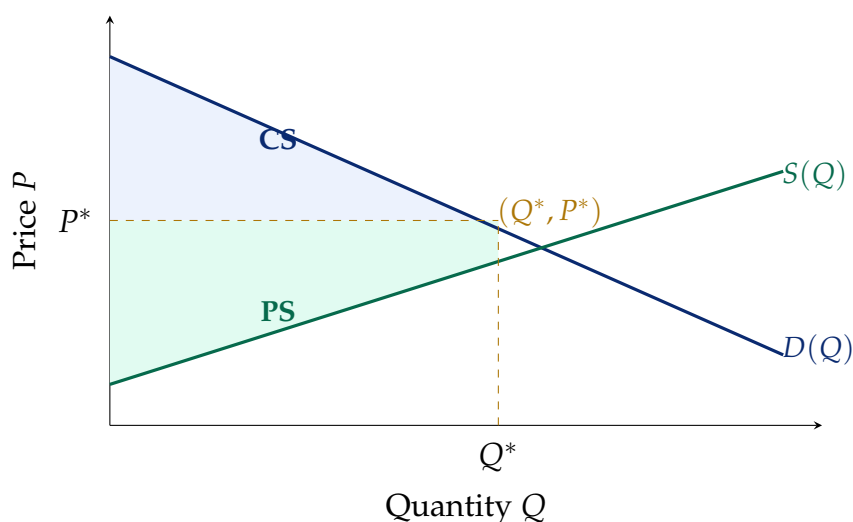
Consumer Surplus (CS) is the area below the demand curve and above the equilibrium price:

$$CS = \int_0^{Q^*} [D(Q) - P^*] dQ$$

Producer Surplus (PS) is the area above the supply curve and below the equilibrium price:

$$PS = \int_0^{Q^*} [P^* - S(Q)] dQ$$

Total surplus = CS + PS. **Deadweight Loss (DWL)** from a distortion is the reduction in total surplus. (*Chiang & Wainwright, p. 421; Hoy et al., p. 351*)



Consumer surplus (blue) and producer surplus (green) at the competitive equilibrium.

Worked Example Computing Consumer and Producer Surplus

Market: $D(Q) = 10 - Q$ (inverse demand), $S(Q) = 2 + Q$ (inverse supply).

Equilibrium: $10 - Q = 2 + Q \Rightarrow Q^* = 4, P^* = 6$.

Consumer Surplus:

$$CS = \int_0^4 [(10 - Q) - 6] dQ = \int_0^4 (4 - Q) dQ = \left(4Q - \frac{Q^2}{2}\right) \Big|_0^4 = 16 - 8 = 8$$

Producer Surplus:

$$PS = \int_0^4 [6 - (2 + Q)] dQ = \int_0^4 (4 - Q) dQ = 8$$

Total surplus = 16.

Effect of a price floor at $\bar{P} = 7$:

At $\bar{P} = 7$: $Q_d = 10 - 7 = 3$, $Q_s = 7 - 2 = 5$. Only $Q_d = 3$ units traded (demand is binding).

$$CS' = \int_0^3 [(10 - Q) - 7] dQ = \int_0^3 (3 - Q) dQ = \left(3Q - \frac{Q^2}{2}\right) \Big|_0^3 = 9 - 4.5 = 4.5$$

$$PS' = \int_0^3 [7 - (2 + Q)] dQ = \int_0^3 (5 - Q) dQ = 15 - 4.5 = 10.5$$

Total surplus = 15. DWL = $16 - 15 = 1$.

4.2 Present Value of Continuous Income Streams

Definition Present Value of a Continuous Stream

The **present value** at time $t = 0$ of a continuous income stream $f(t)$ over $[0, T]$, discounted at continuous rate r :

$$PV = \int_0^T f(t) e^{-rt} dt$$

For a perpetual stream ($T \rightarrow \infty$): $PV = \int_0^\infty f(t) e^{-rt} dt$, provided the integral converges.

Worked Example Present Value Calculations

1. Constant stream: $f(t) = C$, discount rate r , horizon T :

$$PV = \int_0^T C e^{-rt} dt = -\frac{C}{r} e^{-rt} \Big|_0^T = \frac{C}{r} (1 - e^{-rT})$$

As $T \rightarrow \infty$: $PV \rightarrow C/r$ (perpetuity formula).

2. Growing stream: $f(t) = A e^{gt}$, horizon T , $r \neq g$:

$$PV = A \int_0^T e^{(g-r)t} dt = \frac{A}{g-r} (e^{(g-r)T} - 1) = \frac{A}{r-g} (1 - e^{(g-r)T})$$

3. Investment project: revenues $R(t) = 1000e^{0.02t}$, costs $c = 400$ /year, horizon $T = 5$, $r = 0.08$:

$$\begin{aligned} NPV &= \int_0^5 (1000e^{0.02t} - 400)e^{-0.08t} dt = 1000 \int_0^5 e^{-0.06t} dt - 400 \int_0^5 e^{-0.08t} dt \\ &= \frac{1000}{0.06} (1 - e^{-0.3}) - \frac{400}{0.08} (1 - e^{-0.4}) \approx 16667(0.2592) - 5000(0.3297) \approx 4320 - 1649 = 2671 \end{aligned}$$

Since $NPV > 0$, the project is worth undertaking.

4. Lorenz Curve and Gini Coefficient: The Lorenz curve $L(s)$ gives the share of income held by the bottom s fraction of the population. The Gini coefficient is:

$$G = 1 - 2 \int_0^1 L(s) ds$$

If $L(s) = s^2$ (quadratic Lorenz curve): $\int_0^1 s^2 ds = 1/3$, so $G = 1 - 2/3 = 1/3$.

4.3 Capital Accumulation**Worked Example Net Investment and Capital Stock**

Net investment is $I(t) = \dot{K}(t)$ (the flow of new capital). The capital stock at time T is:

$$K(T) = K(0) + \int_0^T I(t) dt$$

Example: $I(t) = 10\sqrt{t}$ with $K(0) = 50$:

$$K(T) = 50 + \int_0^T 10t^{1/2} dt = 50 + \frac{20t^{3/2}}{3} \Big|_0^T = 50 + \frac{20T^{3/2}}{3}$$

At $T = 9$: $K(9) = 50 + 20(27)/3 = 50 + 180 = 230$.

5 First-Order Differential Equations

5.1 Definitions and Classification

Definition Differential Equation

A **differential equation** is an equation involving a function and its derivatives. A **first-order ordinary differential equation** (ODE) has the form:

$$\frac{dy}{dt} = f(t, y) \quad \text{or equivalently} \quad F(t, y, \dot{y}) = 0$$

The **general solution** contains an arbitrary constant C ; an **initial condition** $y(t_0) = y_0$ pins down the **particular solution**.

Definition Autonomous vs. Non-Autonomous

A differential equation $\dot{y} = f(t, y)$ is:

- **Autonomous:** $\dot{y} = f(y)$ — the right-hand side does not explicitly depend on t . The dynamics depend only on the current state, not on time itself.
- **Non-autonomous:** $\dot{y} = f(t, y)$ — time appears explicitly.

Autonomous equations arise naturally in steady-state economic models (Solow, Ramsey). Their phase diagrams are one-dimensional and greatly simplify analysis.

5.2 Linear First-Order ODEs

Theorem Solution of the Linear First-Order ODE

The linear first-order ODE:

$$\dot{y} + a(t)y = b(t)$$

has the general solution (via the **integrating factor** $\mu(t) = e^{\int a(t) dt}$):

$$y(t) = e^{-\int a(t) dt} \left[\int b(t) e^{\int a(t) dt} dt + C \right]$$

Constant-coefficient case ($a(t) = a$, $b(t) = b$):

$$y(t) = \underbrace{Ae^{-at}}_{\text{complementary (homogeneous)}} + \underbrace{\frac{b}{a}}_{\text{particular solution}} \quad A = y_0 - \frac{b}{a}$$

The **equilibrium** (particular solution) is $y^* = b/a$; the complementary function Ae^{-at} captures departure from equilibrium. (*Chiang & Wainwright*, p. 468; *Hoy et al.*, p. 577)

Derivation via Integrating Factor. For $\dot{y} + ay = b$, multiply both sides by e^{at} :

$$e^{at}\dot{y} + ae^{at}y = be^{at} \iff \frac{d}{dt}[e^{at}y] = be^{at}$$

Integrating both sides: $e^{at}y = \frac{b}{a}e^{at} + C \Rightarrow y = \frac{b}{a} + Ce^{-at}$.

Applying $y(0) = y_0$: $C = y_0 - b/a = A$. □

Stability of the Constant-Coefficient Linear ODE

For $\dot{y} + ay = b$ with solution $y(t) = Ae^{-at} + b/a$:

- **Stable** ($a > 0$): as $t \rightarrow \infty$, $e^{-at} \rightarrow 0$, so $y(t) \rightarrow y^* = b/a$ regardless of initial conditions. The equilibrium is globally asymptotically stable.
- **Unstable** ($a < 0$): $e^{-at} \rightarrow \infty$, so $y(t)$ diverges from y^* (except from y^* itself).
- **Neutral** ($a = 0$): $y(t) = y_0 + bt$ (linear trend; no equilibrium unless $b = 0$).

The sign of a — the coefficient of y — governs stability.

Worked Example Price Adjustment Dynamics

Suppose the rate of price adjustment is proportional to excess demand:

$$\dot{P} = \alpha[D(P) - S(P)], \quad \alpha > 0$$

For linear demand and supply: $D(P) = a - bP$ and $S(P) = c + dP$:

$$\dot{P} = \alpha[(a - c) - (b + d)P]$$

This is $\dot{P} + \alpha(b + d)P = \alpha(a - c)$, a linear first-order ODE with $a_{\text{ode}} = \alpha(b + d) > 0$ and $b_{\text{ode}} = \alpha(a - c)$.

Solution: $P(t) = [P(0) - P^*]e^{-\alpha(b+d)t} + P^*$, where $P^* = (a - c)/(b + d)$.

Since $\alpha(b + d) > 0$, the equilibrium price P^* is **globally stable**: $P(t) \rightarrow P^*$ as $t \rightarrow \infty$ from any initial condition $P(0)$.

The speed of convergence is $\alpha(b + d)$ — faster adjustment α , steeper demand slope b , or steeper supply slope d all accelerate convergence to equilibrium.

Worked Example The Solow Growth Model

The Solow model in intensive form (per-worker capital $k = K/L$):

$$\dot{k} = sf(k) - (n + \delta)k$$

where $s \in (0, 1)$ is the savings rate, $f(k) = k^\alpha$ is Cobb-Douglas production per worker, n is population growth, and δ is depreciation.

This is a **nonlinear** autonomous ODE (because $f(k) = k^\alpha$ is nonlinear). We cannot solve it in closed form for general α . Instead:

Steady state: $\dot{k} = 0 \Rightarrow sf(k^*) = (n + \delta)k^*$. For $f(k) = k^\alpha$: $sk^{*\alpha} = (n + \delta)k^*$, giving:

$$k^* = \left(\frac{s}{n + \delta} \right)^{1/(1-\alpha)}$$

Linearisation around k^* : Let $\tilde{k} = k - k^*$ (small deviation).

$$\dot{\tilde{k}} \approx [sf'(k^*) - (n + \delta)] \tilde{k}$$

Stability coefficient: $\lambda = sf'(k^*) - (n + \delta) = s\alpha k^{*\alpha-1} - (n + \delta)$.

At the steady state, $sk^{*\alpha} = (n + \delta)k^*$ implies $sk^{*\alpha-1} = (n + \delta)$. Therefore,

$$\lambda = s\alpha k^{*\alpha-1} - (n + \delta) = \alpha(n + \delta) - (n + \delta) = -(1 - \alpha)(n + \delta) < 0.$$

Since $\lambda < 0$ (as $\alpha \in (0, 1)$ and $n + \delta > 0$), the steady state k^* is **locally asymptotically stable**. Capital converges to k^* from any nearby initial condition.

Speed of convergence: $|\lambda| = (1 - \alpha)(n + \delta)$. For typical values $\alpha = 1/3$, $n = 0.01$, $\delta = 0.05$: $|\lambda| = (2/3)(0.06) = 0.04$ — the economy closes 4% of the gap toward k^* per year, implying a half-life of $\ln 2 / 0.04 \approx 17$ years.

5.3 Separable Differential Equations

Definition Separable ODE

A first-order ODE is **separable** if it can be written as:

$$\frac{dy}{dt} = g(t) h(y)$$

Solution method: separate variables and integrate:

$$\int \frac{dy}{h(y)} = \int g(t) dt$$

Worked Example Separable ODEs in Economics

1. Solve $\dot{y} = ry$ (exponential growth/decay):

$$\frac{dy}{y} = r dt \implies \ln |y| = rt + C_0 \implies y(t) = Ae^{rt}$$

Initial condition $y(0) = y_0$: $A = y_0$. So $y(t) = y_0 e^{rt}$.

2. **Logistic growth** (bounded population or market saturation):

$$\dot{N} = rN \left(1 - \frac{N}{K}\right)$$

Separate: $\int \frac{dN}{N(1 - N/K)} = \int r dt$

Using partial fractions: $\frac{1}{N(1 - N/K)} = \frac{1}{N} + \frac{1/K}{1 - N/K}$:

$$\ln N - \ln(1 - N/K) = rt + C$$

$$\ln \frac{N}{1 - N/K} = rt + C \implies \frac{N}{K - N} = Ae^{rt}$$

Solving: $N(t) = \frac{K}{1 + Be^{-rt}}$, $B = \frac{K - N_0}{N_0}$.

As $t \rightarrow \infty$: $N(t) \rightarrow K$ — the carrying capacity. This model fits the adoption of new technologies, market penetration, and population dynamics.

3. Bernoulli equation: $\dot{k} = sk^\alpha - (n + \delta)k$

This is the Solow equation. With $\alpha = 1/2$, let $z = k^{1/2}$, so

$$\dot{z} = \frac{\dot{k}}{2k^{1/2}} = \frac{s\sqrt{k} - (n + \delta)k}{2\sqrt{k}} = \frac{s}{2} - \frac{n + \delta}{2}z.$$

Thus, for $\alpha = 1/2$, the Solow equation becomes a linear first-order ODE in $z = \sqrt{k}$. More generally, Bernoulli equations can be linearised by a suitable power substitution.

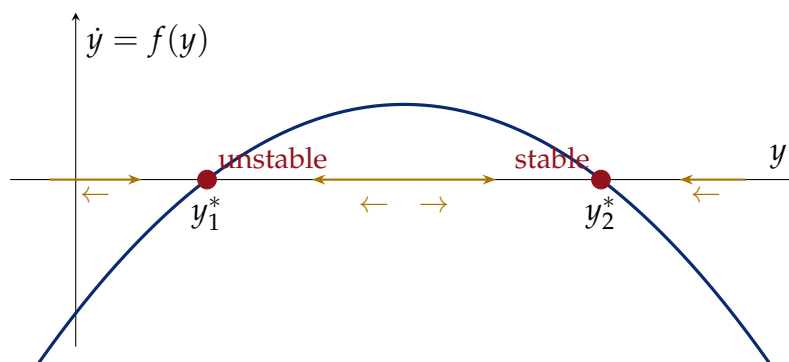
6 Phase Diagrams and Stability Analysis

6.1 Phase Diagrams for Autonomous Equations

For an autonomous ODE $\dot{y} = f(y)$, the **phase diagram** (or phase line) plots $\dot{y}$ against y and allows us to read off the dynamics without solving the equation explicitly.

Reading a Phase Diagram

1. **Equilibria (fixed points):** where $\dot{y} = f(y) = 0$.
2. **Direction of motion:**
 - $f(y) > 0 \Rightarrow \dot{y} > 0 \Rightarrow y$ is *increasing* ($\rightarrow$)
 - $f(y) < 0 \Rightarrow \dot{y} < 0 \Rightarrow y$ is *decreasing* ($\leftarrow$)
3. **Stability of equilibrium y^* :**
 - **Stable** (attractor): arrows point *toward* y^* on both sides, i.e. $f'(y^*) < 0$.
 - **Unstable** (repeller): arrows point *away* from y^* on both sides, i.e. $f'(y^*) > 0$.
 - **Semi-stable**: arrows point toward y^* on one side only.



Phase diagram for $\dot{y} = (y - y_1^*)(y_2^* - y)$. y_1^* is unstable ($f' > 0$); y_2^* is stable ($f' < 0$).

Worked Example Phase Diagram: The Solow Model

The Solow equation $\dot{k} = sk^\alpha - (n + \delta)k$ defines a curve in the $(k, \dot{k})$ plane. Setting $\dot{k} = 0$:

$$sk^\alpha = (n + \delta)k \implies k^* = \left(\frac{s}{n + \delta} \right)^{1/(1-\alpha)}$$

The slope of $\dot{k}$ with respect to k at k^* :

$$\left. \frac{d\dot{k}}{dk} \right|_{k^*} = \alpha sk^{*\alpha-1} - (n + \delta) = \alpha(n + \delta) - (n + \delta) = -(1 - \alpha)(n + \delta) < 0$$

Since the slope is negative at k^* , the equilibrium is **stable**.

Phase-line analysis:

- For $0 < k < k^*$: $sk^\alpha > (n + \delta)k$, so $\dot{k} > 0$ — capital per worker rises.
- For $k > k^*$: $sk^\alpha < (n + \delta)k$, so $\dot{k} < 0$ — capital per worker falls.

From any $k_0 > 0$, the economy converges to k^* .

6.2 Stability Conditions**Theorem Local Stability via Linearisation**

For $\dot{y} = f(y)$ with equilibrium $f(y^*) = 0$, the linearised equation near y^* is:

$$\dot{\tilde{y}} = f'(y^*) \tilde{y}, \quad \tilde{y} = y - y^*$$

Solution: $\tilde{y}(t) = \tilde{y}(0) e^{f'(y^*)t}$.

- **Locally asymptotically stable:** $f'(y^*) < 0$ (eigenvalue negative).
- **Locally unstable:** $f'(y^*) > 0$.
- **Non-hyperbolic** ($f'(y^*) = 0$): linearisation is inconclusive.

(Chiang & Wainwright, p. 475; Hoy et al., p. 584)

Theorem Global Stability (Lyapunov-type)

For $\dot{y} = f(y)$ on $(0, \infty)$ with unique equilibrium y^* : if $f(y) > 0$ for $y < y^*$ and $f(y) < 0$ for $y > y^*$ (phase-line arrows both point toward y^*), then y^* is **globally asymptotically stable**.

7 First-Order Linear Difference Equations

7.1 Definition and Solution

While differential equations model continuous-time dynamics, **difference equations** model discrete-time processes.

Definition First-Order Linear Difference Equation

A first-order linear difference equation with constant coefficients:

$$y_{t+1} = ay_t + b, \quad t = 0, 1, 2, \dots$$

where $a \neq 0$ is the coefficient and b is the constant term.

Theorem Solution of the First-Order Linear Difference Equation

The general solution of $y_{t+1} = ay_t + b$ is:

$$y_t = Aa^t + y^*, \quad y^* = \frac{b}{1-a} \quad (a \neq 1)$$

where $A = y_0 - y^*$ is determined by the initial condition y_0 , and $y^* = b/(1-a)$ is the **intertemporal equilibrium**.

If $a = 1$: $y_t = y_0 + bt$ (linear trend, no equilibrium unless $b = 0$). (Chiang & Wainwright, p. 558; Hoy et al., p. 614)

Stability of the First-Order Difference Equation

The equilibrium $y^* = b/(1-a)$ is:

Condition on a	Behaviour of y_t
$ a < 1$	Stable: $y_t \rightarrow y^*$ monotonically ($a > 0$) or with oscillations ($-1 < a < 0$)
$a = 1$	Neutral: diverges (unless $b = 0$)
$a = -1$	Neutral: permanent oscillations around y^*
$a > 1$	Unstable: explosive growth
$a < -1$	Unstable: explosive oscillations

Worked Example Cobweb Model

In agriculture, supply responds to *last period's* price (production takes time). With linear demand and supply:

$$Q_t^d = \alpha - \beta P_t, \quad Q_t^s = -\gamma + \delta P_{t-1}$$

Market clearing $Q_t^d = Q_t^s$:

$$\alpha - \beta P_t = -\gamma + \delta P_{t-1} \implies P_t = \frac{\alpha + \gamma}{\beta} - \frac{\delta}{\beta} P_{t-1}$$

This is $P_t = aP_{t-1} + b$ with $a = -\delta/\beta$ and $b = (\alpha + \gamma)/\beta$.

Equilibrium: $P^* = (\alpha + \gamma)/(\beta + \delta)$ (same as static equilibrium).

Stability condition: $|a| = \delta/\beta < 1 \iff \delta < \beta$ (supply slope less steep than demand slope in absolute value).

- $\delta < \beta$ ($|a| < 1$): **converging oscillations** — the cobweb spirals in.
- $\delta = \beta$ ($|a| = 1$): **permanent oscillations** — the cobweb circles.
- $\delta > \beta$ ($|a| > 1$): **explosive oscillations** — the cobweb spirals out.

Numerical example: $\alpha = 100, \beta = 2, \gamma = 10, \delta = 1, P_0 = 30$.

Here $b = (\alpha + \gamma)/\beta = (100 + 10)/2 = 55$ and $a = -\delta/\beta = -0.5$. Hence

$$P^* = \frac{b}{1-a} = \frac{55}{1.5} \approx 36.67.$$

$$P_t = (P_0 - P^*)(-0.5)^t + P^* = (30 - 36.67)(-0.5)^t + 36.67 = -6.67(-0.5)^t + 36.67$$

Since $|a| = 0.5 < 1$: **stable** with diminishing oscillations. $P_1 = 55 - 0.5(30) = 40$; $P_2 = 55 - 0.5(40) = 35$; $P_3 = 55 - 0.5(35) = 37.5 \dots$ converging to $P^* \approx 36.67$.

Worked Example National Income Dynamics (Samuelson Multiplier-Accelerator)

The Samuelson (1939) model combines the multiplier and accelerator:

$$Y_t = C_t + I_t + G$$

$$C_t = cY_{t-1}, \quad I_t = b(C_t - C_{t-1}) = bc(Y_{t-1} - Y_{t-2})$$

This leads to a second-order difference equation:

$$Y_t = c(1+b)Y_{t-1} - bcY_{t-2} + G$$

The stability of this system depends on the roots of the characteristic equation $r^2 - c(1+b)r + bc = 0$. The system is stable iff both roots lie within the unit circle $|r| < 1$, which requires (among other conditions) $bc < 1$.

Economic insight: if the accelerator b is large, investment amplifies fluctuations and the economy can exhibit endogenous business cycles — oscillating income even without external shocks.

8 Economic Dynamics: Selected Applications

8.1 The Domar Growth Model

Worked Example Domar's Capacity Growth Condition

Domar (1946) asks: at what rate must investment grow to keep capacity utilisation constant?

Two sides of the economy:

- **Demand side** (Keynesian multiplier): $Y = I/s$, so $\dot{Y} = \dot{I}/s$.
- **Supply side** (capacity): new investment creates capacity. If each unit of investment yields capacity σ , then $\dot{Y}_{\text{capacity}} = \sigma I$.

For full capacity utilisation: $\dot{Y} = \dot{Y}_{\text{capacity}}$:

$$\frac{\dot{I}}{s} = \sigma I \implies \frac{\dot{I}}{I} = s\sigma$$

This is the ODE $\dot{I} = s\sigma I$ with solution $I(t) = I_0 e^{s\sigma t}$.

Domar condition: investment must grow at the warranted rate $g^* = s\sigma$ (savings rate times output-capital ratio) to maintain full employment. This is the foundation of Harrod-Domar growth theory.

8.2 Continuous-Time Discounting and Optimal Control Preview

Worked Example Ramsey Euler Equation

In the Ramsey-Cass-Koopmans model, a representative household maximises:

$$\int_0^\infty U(c(t)) e^{-\rho t} dt$$

subject to $\dot{a} = ra - c$ (asset accumulation), where c is consumption, a is assets, r is the interest rate, and ρ is the discount rate.

The **Euler equation** (from the Hamiltonian / calculus of variations):

$$\frac{\dot{c}}{c} = \frac{r - \rho}{\sigma(c)}$$

where $\sigma(c) = -U''(c)c/U'(c) > 0$ is the elasticity of intertemporal substitution. For CRRA utility $U(c) = c^{1-\sigma}/(1-\sigma)$: $\sigma(c) = \sigma$ (constant), and:

$$\frac{\dot{c}}{c} = \frac{r - \rho}{\sigma}$$

This is a separable ODE: $c(t) = c_0 e^{(r-\rho)t/\sigma}$.

Interpretation: consumption grows when the return to saving (r) exceeds the discount rate (ρ). The larger σ (more patient households), the slower consumption grows for a given $r - \rho$.

9 Practice Questions

9.1 Section A: Integration Techniques

A1. (Indefinite Integrals) Evaluate each integral:

(a) $\int (6x^5 - 4x^3 + 3x^2 - 7) dx$

(b) $\int \frac{2x + 1}{\sqrt{x^2 + x - 3}} dx$

(c) $\int x^2 e^{3x} dx$

(d) $\int \frac{3x^2 - 2}{x^3 - 2x + 5} dx$

(e) $\int \frac{\ln x}{x} dx, \quad x > 0$

A2. (Definite Integrals) Evaluate:

(a) $\int_1^4 \frac{3}{\sqrt{x}} dx$

(b) $\int_0^1 x e^{-x} dx$

(c) $\int_0^\infty e^{-2t} dt$

(d) $\int_1^e \ln x dx$

(e) $\int_0^2 (3x^2 - 2x + 1) dx$

A3. (Consumer and Producer Surplus) The market for a good has:

$$D(Q) = 50 - 2Q \quad (\text{inverse demand}), \quad S(Q) = 10 + 3Q \quad (\text{inverse supply})$$

(a) Find the competitive equilibrium (Q^*, P^*) .

(b) Compute CS, PS, and total surplus.

(c) A per-unit tax of $t = 5$ is imposed on producers (supply shifts up by 5). Find the new equilibrium, CS, PS, tax revenue, and DWL.

(d) Verify: $DWL = \text{loss in total surplus} - \text{tax revenue}$.

A4. (Present Value)

- (a) Find the present value of a constant income stream of \$800/year for 10 years at a continuous discount rate of 5%.
- (b) A project yields revenues $R(t) = 2000e^{0.03t}$ continuously for $T = 8$ years. At a discount rate of $r = 0.10$, compute the present value of revenues.
- (c) Find the present value of a perpetual payment of \$500/year growing at 2% per year, discounted at 6%.

9.2 Section B: Differential Equations**B1. (Linear First-Order ODEs) Solve each ODE:**

- (a) $\dot{y} + 3y = 12$, $y(0) = 5$
- (b) $\dot{y} - 2y = 6e^t$, $y(0) = 1$
- (c) $\dot{P} + 0.5P = 20$, $P(0) = 30$ (price dynamics)
- (d) $\dot{k} + (n + \delta)k = sf(k)$ with $f(k) = Ak$ (AK model), $k(0) = k_0$. Solve and find the steady-state growth rate of k .

B2. (Separable ODEs) Solve:

- (a) $\dot{y} = y^2$, $y(0) = 1$
- (b) $\dot{y} = -ky$, $y(0) = Y_0$ (radioactive decay / depreciation)
- (c) $\dot{N} = rN(1 - N/K)$, $N(0) = N_0 < K$ (logistic growth)
- (d) $\frac{dy}{dt} = \frac{t}{y}$, $y(0) = 2$

B3. (Phase Diagrams and Stability) For each autonomous ODE: (i) find all equilibria; (ii) draw a phase diagram; (iii) classify each equilibrium as stable or unstable; (iv) find $y(t)$ if $y(0) = y_0$ is given.

- (a) $\dot{y} = (3 - y)(y - 1)$
- (b) $\dot{k} = 0.4k^{0.5} - 0.06k$, $k(0) = 2$
- (c) $\dot{y} = y(2 - y) - 1$

B4. (Economic Dynamics)

- (a) **(Solow Model)** For $\dot{k} = 0.3k^{1/3} - (0.05 + 0.02)k$ with $k(0) = 4$: find k^* , linearise around k^* , find the speed of convergence, and compute the approximate half-life.
- (b) **(Price Dynamics)** Demand $D(P) = 120 - 4P$, supply $S(P) = 20 + 2P$, and adjustment $\dot{P} = 0.5[D(P) - S(P)]$. Solve for $P(t)$ with $P(0) = 25$. Is the market stable?

- (c) **(Domar Model)** If $s = 0.2$ and $\sigma = 0.5$ (output-capital ratio), find the warranted growth rate. Compute $I(t)$ if $I(0) = 100$. In how many years does investment double?

9.3 Section C: Difference Equations

C1. (Solving Difference Equations) Solve each difference equation:

- (a) $y_{t+1} = 0.6y_t + 8, \quad y_0 = 5$
- (b) $y_{t+1} = 2y_t - 10, \quad y_0 = 7$
- (c) $P_t = -0.5P_{t-1} + 45, \quad P_0 = 60$ (cobweb model)
- (d) $y_{t+1} = -y_t + 6, \quad y_0 = 0$

C2. (Cobweb Model) A market has demand $Q^d = 100 - 2P_t$ and supply $Q^s = -10 + 3P_{t-1}$ (supply depends on last period's price).

- (a) Derive the difference equation for P_t .
- (b) Find the equilibrium price P^* .
- (c) Solve for P_t with $P_0 = 15$.
- (d) Is the market stable? What is the speed of convergence?
- (e) Interpret: what does stability mean for actual market behaviour?

10 Full Solutions

10.1 Solutions: Section A

Solution

A1. Indefinite Integrals

(a) $\int (6x^5 - 4x^3 + 3x^2 - 7) dx = x^6 - x^4 + x^3 - 7x + C$

(b) Notice $\frac{d}{dx}(x^2 + x - 3) = 2x + 1$. Let $u = x^2 + x - 3$, $du = (2x + 1)dx$:

$$\int \frac{du}{\sqrt{u}} = 2\sqrt{u} + C = 2\sqrt{x^2 + x - 3} + C$$

(c) Integration by parts twice.

Round 1: $u = x^2$, $dv = e^{3x}dx$; $du = 2x dx$, $v = e^{3x}/3$:

$$\int x^2 e^{3x} dx = \frac{x^2 e^{3x}}{3} - \frac{2}{3} \int x e^{3x} dx$$

Round 2: $u = x$, $dv = e^{3x}dx$; $du = dx$, $v = e^{3x}/3$:

$$\int x e^{3x} dx = \frac{x e^{3x}}{3} - \frac{e^{3x}}{9} + C$$

Combining:

$$\int x^2 e^{3x} dx = \frac{x^2 e^{3x}}{3} - \frac{2}{3} \left(\frac{x e^{3x}}{3} - \frac{e^{3x}}{9} \right) + C = e^{3x} \left(\frac{x^2}{3} - \frac{2x}{9} + \frac{2}{27} \right) + C$$

(d) Numerator is the derivative of denominator: let $u = x^3 - 2x + 5$, $du = (3x^2 - 2)dx$: $\int \frac{du}{u} = \ln |x^3 - 2x + 5| + C$

(e) Let $u = \ln x$, $du = dx/x$: $\int \frac{\ln x}{x} dx = \int u du = \frac{u^2}{2} + C = \frac{(\ln x)^2}{2} + C$

Solution

A2. Definite Integrals

(a) $\int_1^4 3x^{-1/2} dx = 6x^{1/2} \Big|_1^4 = 6(2) - 6(1) = 12 - 6 = 6$

(b) Integration by parts: $u = x$, $dv = e^{-x}dx$; $du = dx$, $v = -e^{-x}$:

$$\int_0^1 x e^{-x} dx = -x e^{-x} \Big|_0^1 + \int_0^1 e^{-x} dx = -e^{-1} + (-e^{-x}) \Big|_0^1 = -e^{-1} + (-e^{-1} + 1) = 1 - 2e^{-1} \approx 0.264$$

(c) $\int_0^\infty e^{-2t} dt = -\frac{e^{-2t}}{2} \Big|_0^\infty = 0 - \left(-\frac{1}{2}\right) = \frac{1}{2}$

(d) $\int_1^e \ln x dx = (x \ln x - x) \Big|_1^e = (e \cdot 1 - e) - (1 \cdot 0 - 1) = 0 + 1 = 1$

(e) $\int_0^2 (3x^2 - 2x + 1) dx = (x^3 - x^2 + x) \Big|_0^2 = (8 - 4 + 2) - 0 = 6$

Solution**A3. Consumer and Producer Surplus**

$$D(Q) = 50 - 2Q, S(Q) = 10 + 3Q.$$

(a) Equilibrium: $50 - 2Q = 10 + 3Q \Rightarrow 5Q = 40 \Rightarrow Q^* = 8, P^* = 34.$

(b) $CS = \int_0^8 [(50 - 2Q) - 34] dQ = \int_0^8 (16 - 2Q) dQ = (16Q - Q^2) \Big|_0^8 = 128 - 64 = 64$

$$PS = \int_0^8 [34 - (10 + 3Q)] dQ = \int_0^8 (24 - 3Q) dQ = (24Q - \frac{3Q^2}{2}) \Big|_0^8 = 192 - 96 = 96$$

$$\text{Total surplus} = 64 + 96 = 160.$$

(c) With tax $t = 5$ on producers, supply becomes $S'(Q) = 15 + 3Q.$

New equilibrium: $50 - 2Q = 15 + 3Q \Rightarrow Q^{**} = 7, P_d^{**} = 36$ (consumer price), $P_s^{**} = P_d^{**} - t = 31$ (producer price).

$$CS' = \int_0^7 [(50 - 2Q) - 36] dQ = \int_0^7 (14 - 2Q) dQ = (14Q - Q^2) \Big|_0^7 = 98 - 49 = 49$$

$$PS' = \int_0^7 [31 - (10 + 3Q)] dQ = \int_0^7 (21 - 3Q) dQ = (21Q - \frac{3Q^2}{2}) \Big|_0^7 = 147 - 73.5 = 73.5$$

$$\text{Tax revenue} = t \cdot Q^{**} = 5 \times 7 = 35.$$

$$\text{Total surplus after tax} = 49 + 73.5 + 35 = 157.5.$$

$$DWL = 160 - 157.5 = 2.5.$$

(d) **Verification:** Loss in total surplus $= 160 - 157.5 = 2.5$. Tax revenue $= 35$.
 $DWL = \text{loss in total surplus} = 2.5$ (not loss minus tax revenue; the tax is a transfer, not a deadweight loss). The DWL is the pure efficiency loss from the reduced quantity $Q^* - Q^{**} = 1$ unit. ✓

Solution**A4. Present Value**

(a) $PV = \int_0^{10} 800e^{-0.05t} dt = -\frac{800}{0.05}e^{-0.05t} \Big|_0^{10} = 16000(1 - e^{-0.5}) \approx 16000(0.3935) \approx \$6,296$

(b) $PV = \int_0^8 2000e^{0.03t} \cdot e^{-0.10t} dt = 2000 \int_0^8 e^{-0.07t} dt = \frac{2000}{0.07}(1 - e^{-0.56}) \approx 28571(0.4288) \approx \$12,252$

(c) Growing perpetuity: $f(t) = 500e^{0.02t}, r = 0.06: PV = \int_0^{\infty} 500e^{0.02t}e^{-0.06t} dt = 500 \int_0^{\infty} e^{-0.04t} dt = \frac{500}{0.04} = \$12,500$

10.2 Solutions: Section B

Solution**B1. Linear First-Order ODEs**

(a) $\dot{y} + 3y = 12$: equilibrium $y^* = 12/3 = 4$.

$$y(t) = (y_0 - y^*)e^{-3t} + y^* = (5 - 4)e^{-3t} + 4 = e^{-3t} + 4$$

Since $a = 3 > 0$: $y(t) \rightarrow 4$ as $t \rightarrow \infty$. Stable.

(b) $\dot{y} - 2y = 6e^t$: integrating factor $\mu = e^{-2t}$.

$$\frac{d}{dt}[e^{-2t}y] = 6e^{-2t} \cdot e^t = 6e^{-t}$$

$$e^{-2t}y = -6e^{-t} + C \Rightarrow y = -6e^t + Ce^{2t}$$

$$y(0) = 1: -6 + C = 1 \Rightarrow C = 7.$$

$$y(t) = -6e^t + 7e^{2t}. \text{ (Unstable: diverges as } t \rightarrow \infty.)$$

(c) $\dot{P} + 0.5P = 20$: $P^* = 20/0.5 = 40$.

$$P(t) = (30 - 40)e^{-0.5t} + 40 = -10e^{-0.5t} + 40$$

Stable: $P(t) \rightarrow 40$ as $t \rightarrow \infty$.

(d) AK model: $\dot{k} + (n + \delta)k = sAk \Rightarrow \dot{k} = [sA - (n + \delta)]k$

This is $\dot{k} = \lambda k$ with $\lambda = sA - (n + \delta)$.

$$k(t) = k_0 e^{[sA - (n + \delta)]t}$$

Steady-state growth rate: $g_k = sA - (n + \delta)$. Unlike the Solow model with diminishing returns, the AK model has constant returns to capital, so there is no steady state level of k (unless $g_k = 0$) — the economy grows permanently at rate g_k if $sA > n + \delta$.

Solution**B2. Separable ODEs**

(a) $\dot{y} = y^2$: $\int y^{-2} dy = \int dt \Rightarrow -1/y = t + C$.

$$y(0) = 1: -1 = C. \text{ So } -1/y = t - 1 \Rightarrow y(t) = \frac{1}{1 - t}.$$

Blows up at $t = 1$ (**finite escape time**).

(b) $\dot{y} = -ky$: $\ln|y| = -kt + C_0 \Rightarrow y(t) = Y_0 e^{-kt}$.

Exponential decay; $y(t) \rightarrow 0$ as $t \rightarrow \infty$.

(c) Logistic growth: $\int \frac{dN}{N(1 - N/K)} = \int r dt$

$$\text{Partial fractions: } \frac{1}{N(1 - N/K)} = \frac{1}{N} + \frac{1}{K - N}$$

$$\ln N - \ln(K - N) = rt + C_0 \Rightarrow \frac{N}{K - N} = Ae^{rt}$$

$$N(0) = N_0: A = N_0/(K - N_0).$$

$$N(t) = \frac{K}{1 + \left(\frac{K - N_0}{N_0}\right)e^{-rt}} \rightarrow K \text{ as } t \rightarrow \infty.$$

(d) $\dot{y} = t/y$: $y dy = t dt \Rightarrow y^2/2 = t^2/2 + C$.

$$y(0) = 2: 4/2 = C, \text{ so } C = 2. \quad y^2 = t^2 + 4 \Rightarrow y(t) = \sqrt{t^2 + 4} \text{ (taking positive root).}$$

Solution**B3. Phase Diagrams and Stability**(a) $\dot{y} = (3 - y)(y - 1)$.Equilibria: $y_1^* = 1$ and $y_2^* = 3$.

$$f'(y) = (y - 1)(-1) + (3 - y)(1) = -y + 1 + 3 - y = 4 - 2y$$

 $f'(1) = 4 - 2 = 2 > 0$: $y_1^* = 1$ is **unstable**. $f'(3) = 4 - 6 = -2 < 0$: $y_2^* = 3$ is **stable**.General solution: this factors as $-(y - 1)(y - 3)$. The ODE is separable:

$$\int \frac{dy}{(y - 1)(y - 3)} = \int -dt.$$

By partial fractions: $\frac{1}{(y - 1)(y - 3)} = \frac{1}{-2} \left(\frac{1}{y - 1} - \frac{1}{y - 3} \right)$:

$$-\frac{1}{2} \ln \left| \frac{y-1}{y-3} \right| = -t + C_0$$

 $\left| \frac{y-1}{y-3} \right| = Ae^{2t}$. As $t \rightarrow \infty$: if $1 < y_0 < 3$, $y(t) \rightarrow 3$ (stable equilibrium). ✓(b) $\dot{k} = 0.4k^{0.5} - 0.06k$.

Equilibrium:

$$0.4k^{0.5} = 0.06k \Rightarrow \frac{0.4}{0.06} = \frac{k}{k^{0.5}} = k^{0.5} \Rightarrow k^{*0.5} = \frac{20}{3} \Rightarrow k^* = \left(\frac{20}{3} \right)^2 \approx 44.4.$$

 $f'(k) = 0.2k^{-0.5} - 0.06$. At k^* : $f'(k^*) = 0.2(3/20) - 0.06 = 0.03 - 0.06 = -0.03 < 0$: **stable**.Speed of convergence: $|\lambda| = 0.03$ (3% per unit time).Starting from $k(0) = 2 < k^*$: since $f(2) = 0.4\sqrt{2} - 0.12 \approx 0.566 - 0.12 = 0.446 > 0$, k is increasing toward k^* .(c) $\dot{y} = y(2 - y) - 1 = 2y - y^2 - 1 = -(y^2 - 2y + 1) = -(y - 1)^2$.Equilibrium: $(y - 1)^2 = 0 \Rightarrow y^* = 1$ (only one equilibrium). $f'(y^*) = -(2y - 2)|_{y=1} = 0$: non-hyperbolic, linearisation inconclusive.Since $(y - 1)^2 \geq 0$, we have $\dot{y} = -(y - 1)^2 \leq 0$ with equality only at $y = 1$. For any $y_0 \neq 1$, $\dot{y} < 0$ — y is always decreasing (or zero). $y^* = 1$ is a **semi-stable** equilibrium: approached from above ($y > 1$) but not below ($y < 1$ continues to decrease away from 1).**Solution****B4. Economic Dynamics**(a) Solow: $\dot{k} = 0.3k^{1/3} - 0.07k$.Steady state: $0.3k^{*1/3} = 0.07k^* \Rightarrow k^{*2/3} = 0.3/0.07 \approx 4.286 \Rightarrow k^* \approx 4.286^{3/2} \approx 8.87$.Linearisation: $\lambda = f'(k^*) = 0.1k^{*-2/3} - 0.07 = 0.1(1/4.286) - 0.07 = 0.02333 - 0.07 = -0.04667$ Speed of convergence: $|\lambda| \approx 4.67\%$ per year.Half-life: $t_{1/2} = \ln 2 / 0.04667 \approx 14.9$ years.(b) $\dot{P} = 0.5[D(P) - S(P)] = 0.5[120 - 4P - 20 - 2P] = 0.5[100 - 6P] = 50 - 3P$.This is $\dot{P} + 3P = 50$. Equilibrium: $P^* = 50/3 \approx 16.67$.

$P(t) = (P_0 - P^*)e^{-3t} + P^* = (25 - 50/3)e^{-3t} + 50/3 = (25/3)e^{-3t} + 50/3$
 Since coefficient $a = 3 > 0$: **stable**. $P(t) \rightarrow 50/3 \approx 16.67$ as $t \rightarrow \infty$.
 Check equilibrium: $D(P^*) = 120 - 4(50/3) = 120 - 200/3 = (360 - 200)/3 = 160/3 \approx 53.3$. $S(P^*) = 20 + 2(50/3) = 20 + 100/3 = (60 + 100)/3 = 160/3$.
 $D = S$. ✓
 (c) $g^* = s\sigma = 0.2 \times 0.5 = 0.10$ (10% warranted growth rate).
 $I(t) = 100e^{0.10t}$.
 Doubling time: $e^{0.10t} = 2 \Rightarrow t = \ln 2 / 0.10 \approx 6.93$ years.

10.3 Solutions: Section C

Solution

C1. Difference Equations

(a) $y_{t+1} = 0.6y_t + 8$: $y^* = 8/(1 - 0.6) = 20$.
 $y_t = (y_0 - y^*)a^t + y^* = (5 - 20)(0.6)^t + 20 = -15(0.6)^t + 20$.
 Since $|a| = 0.6 < 1$: $y_t \rightarrow 20$. Stable with monotonic convergence (since $a > 0$).
 (b) $y_{t+1} = 2y_t - 10$: $y^* = -10/(1 - 2) = 10$.
 $y_t = (7 - 10)(2)^t + 10 = -3(2)^t + 10$.
 Since $|a| = 2 > 1$: unstable. $y_t \rightarrow -\infty$ as $t \rightarrow \infty$.
 (c) $P_t = -0.5P_{t-1} + 45$: $P^* = 45/(1 - (-0.5)) = 45/1.5 = 30$.
 $P_t = (60 - 30)(-0.5)^t + 30 = 30(-0.5)^t + 30$.
 Since $|a| = 0.5 < 1$: stable with oscillating convergence (since $a < 0$).
 $P_1 = 45 - 0.5(60) = 15$; $P_2 = 45 - 0.5(15) = 37.5$; $P_3 = 45 - 0.5(37.5) = 26.25 \dots \rightarrow 30$.
 (d) $y_{t+1} = -y_t + 6$: $y^* = 6/(1 - (-1)) = 3$.
 $y_t = (y_0 - y^*)(-1)^t + y^* = (0 - 3)(-1)^t + 3 = -3(-1)^t + 3$.
 Since $|a| = 1$: neutral stability — permanent oscillations. $y_0 = 0, y_1 = 6, y_2 = 0, y_3 = 6 \dots$ perpetual 2-cycle.

Solution

C2. Cobweb Model

$Q^d = 100 - 2P_t, Q^s = -10 + 3P_{t-1}$.
 (a) Market clearing: $100 - 2P_t = -10 + 3P_{t-1}$
 $P_t = \frac{110 - 3P_{t-1}}{2} = 55 - 1.5P_{t-1}$
 So $a = -1.5$ and $b = 55$.
 (b) $P^* = b/(1 - a) = 55/(1 - (-1.5)) = 55/2.5 = 22$.
 (c) $P_t = (P_0 - P^*)a^t + P^* = (15 - 22)(-1.5)^t + 22 = -7(-1.5)^t + 22$.
 (d) Since $|a| = 1.5 > 1$: the equilibrium is **unstable**.
 Prices oscillate with **increasing amplitude**: $P_1 = 55 - 1.5(15) = 32.5$; $P_2 = 55 - 1.5(32.5) = 6.25$; $P_3 = 55 - 1.5(6.25) = 45.625 \dots$
 The deviations from $P^* = 22$ grow each period by a factor of 1.5.
 (e) **Interpretation**: When $\delta/\beta = 1.5 > 1$ (supply slope exceeds demand slope), small disturbances from equilibrium grow over time. Farmers over-react to current prices when setting production: a high price today causes too much supply next period, crashing the price, which leads to too little supply the period

after, and so on. The market does *not* converge — real-world commodity markets sometimes exhibit this explosive cobweb behaviour, motivating government stabilisation policies (buffer stocks, price support schemes).

Summary: Key Results

Concept	Formula / Condition
Power rule (integral)	$\int x^n dx = x^{n+1}/(n+1) + C, n \neq -1$
Substitution rule	$\int f(g(x))g'(x)dx = \int f(u)du, u = g(x)$
Integration by parts	$\int u dv = uv - \int v du$
FTC (evaluation)	$\int_a^b f(x)dx = F(b) - F(a)$
Consumer Surplus	$CS = \int_0^{Q^*} [D(Q) - P^*]dQ$
PV of stream	$PV = \int_0^T f(t)e^{-rt}dt; \text{perpetuity} = C/r$
Growing perpetuity	$PV = C/(r - g), r > g$
Linear ODE solution	$\dot{y} + ay = b: y(t) = (y_0 - b/a)e^{-at} + b/a$
Stability (ODE)	$a > 0$: stable; $a < 0$: unstable; $f'(y^*) < 0$: locally stable
Separable ODE	$\dot{y} = g(t)h(y): \int dy/h(y) = \int g(t)dt$
Solow steady state	$k^* = (s/(n + \delta))^{1/(1-\alpha)}$ (Cobb-Douglas)
Speed of convergence	$ \lambda = (1 - \alpha)(n + \delta); \text{half-life} = \ln 2/ \lambda $
Difference equation	$y_{t+1} = ay_t + b: y_t = (y_0 - y^*)a^t + y^*, y^* = b/(1 - a)$
Stability (diff. eq.)	$ a < 1$: stable; $ a > 1$: unstable; $ a = 1$: neutral
Cobweb stability	$ \delta/\beta < 1$ (supply slope < demand slope in abs. value)
Logistic growth	$N(t) = K/(1 + Be^{-rt}), B = (K - N_0)/N_0$
Domar growth	$g^* = s\sigma$ (warranted growth rate)

References

- **Primary:** Chiang & Wainwright, *Fundamental Methods of Mathematical Economics*, 4th ed., McGraw-Hill, 2005. Chapters 13–16.
- **Supplementary:** Hoy et al., *Mathematics for Economics*, 3rd ed., MIT Press, 2011. Chapters 14–15, 25–26.
- **Supplementary:** Simon & Blume, *Mathematics for Economists*, Norton, 1994. Chapters 23–24.

End of Notes — ECON 320 Integration and Economic Dynamics